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Design and implementation of a smart earthquake rescue robot to enhance rescue operations

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In the wake of catastrophic earthquakes, rescue operations encounter significant obstacles in locating and reaching individuals trapped under debris. This study introduces the Smart Earthquake Rescue Robot (SERR) prototype, a cutting-edge solution designed to enhance the efficiency and effectiveness of earthquake rescue missions. The SERR is a mobile robot with advanced features, including live video streaming through an integrated camera, Grid-Eye temperature detection, and provisions for communication via built-in speakers and microphones, although these audio communication capabilities are pending implementation in the current prototype. It can be remotely controlled using a smartphone application, offering a safer and more efficient method for conducting rescue operations. Unlike recent advances discussed in the literature, SERR uniquely combines visual, thermal, and audio data with a multi-modal Convolutional Neural Network with Long Short-Term Memory (CNN-LSTM) model (RescueNet), achieving high accuracy (0.94), precision (0.90), recall (0.96), and F1-score (0.92) in detecting survivors, as validated by MATLAB simulations using USGS-PAGER data. The SERR's rapid runtime (35 ms) highlights its promise as a tool to improve earthquake rescue outcomes.

Keywords Smart earthquake rescue robot (SERR), Earthquake rescue operations, Human detection, Thermal imaging, Wireless robotics

On February 6, 2023, two massive earthquakes with magnitudes of 7.7 and 7.5 struck south-eastern Turkey and northwestern Syria. According to the United Nations High Commissioner for Refugees (UNHCR), the earthquake affected an estimated 15.73 million people in Turkey and Syria, killing approximately 55,000 and injuring nearly 130,000. Millions have been displaced from their homes[1]. Rescue operations after earthquakes generally involve many steps. The rescue team uses dogs to enter the area and look for casualties on the surface. Then, it examines the situation beneath the rubble with video cameras. The rescue crew then seeks to confirm whether there are any trapped individuals. The most critical responsibilities in rescue operations are to assess the situation, locate the victims, and establish contact with them. However, human rescuers face great difficulty and risk attempting in this respect[2].

There are several techniques based on sensor technologies in rescue operations. Using specialist sound equipment can pinpoint the slightest disturbances within a few meters. The site should remain silent while a rescue team member hammers thrice to elicit a response[3]. Generally, it is possible to find people buried under rubble using audio signal analysis. However, microphone accuracy suffers when there is a lot of background noise[4]. Another drawback is the inability of auditory detection technologies to locate unconscious individuals. To address these gaps, this paper presents the Smart Earthquake Rescue Robot (SERR) prototype, designed to enhance post-earthquake rescue operations by improving the efficiency and safety of locating survivors. SERR overcomes key limitations of existing sensor-based techniques, such as the susceptibility of audio detection to background noise and the inability to locate unconscious individuals. It offers a versatile, reliable solution that reduces risks to rescue teams through wireless remote control and advanced detection capabilities. SERR integrates live video streaming via an attached camera, thermal detection using a Grid-Eye sensor to identify living humans or animals under rubble, and a user-friendly smartphone application for remote operation. Additionally, a light enhances video stream quality, controllable via the app, while provisions for two-way audio communication (pending implementation) are included for future development.

The robot integrates multiple functionalities to achieve this:

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- Survivor Detection: The robot can detect living humans or animals under rubble using cameras and thermal sensors.
- Obstacle Navigation: The robot has autonomous navigation capabilities to move through chaotic environments safely.
- Remote Control: A smartphone application allows rescue teams to control the robot remotely, ensuring it can be directed precisely where needed.

In order to assess the performance of the SERR, this research integrated data from the USGS—PAGER (Prompt Assessment of Global Earthquakes for Response) system[58]. The PAGER system offers crucial assessments of the impact on fatalities and economic losses resulting from large earthquakes around the world. This dataset contains comprehensive data on the probable magnitude of structural damage to buildings, the probability of casualties, and the estimated financial consequences. These factors are crucial for developing accurate simulation environments. Nevertheless, acknowledging the significance of audio data in identifying human presence amidst debris, a few alterations were made to the PAGER-CAT dataset by incorporating simulated audio samples. The audio files were created to simulate several sounds that could be encountered in a post-earthquake situation, including distress calls, tapping, and other sounds produced by humans. This modification was essential to ensure that the simulations precisely reflect the demanding conditions encountered during real earthquake rescue operations. This multifaceted approach ensures that SERR can effectively navigate complex environments while providing critical live updates and communication capabilities to rescue teams. This Smart Earthquake Rescue Robot (SERR), a prototype uniquely combining live video streaming, Grid-Eye thermal detection for unconscious victims, and provisions for audio communication (pending implementation), surpasses[56,57] by fusing visual, thermal, and audio data via a CNN-LSTM model (RescueNet). Validated through MATLAB simulations with USGS-PAGER data, SERR achieves high accuracy (0.94) and rapid runtime (35 ms), offering a safer, more effective solution for real-time disaster response in telecommunications, smart cities, and autonomous vehicle industries. The rest of the paper is organised as follows: Section II reviews related works. Section III details the design of the proposed SERR. Section IV covers the implementation and testing of the SERR, while Section V presents the performance evaluation. Finally, Section VI provides the conclusion.

Related works

In this section, this research work considers numerous earthquake rescue robots and adopts a systematic approach to ensure the inclusion of the most relevant and high-quality studies in the literature review. The selection process involved the following steps:

Database Search: We conducted comprehensive searches in multiple academic databases, including IEEE Xplore, PubMed, Scopus, and Google Scholar, using keywords such as"earthquake rescue robot,""human detection,""thermal imaging,“and”wireless robotics".

Inclusion Criteria: Papers were selected based on their relevance to the following criteria:

- Focus on robotic systems for earthquake rescue operations.
- Utilisation of advanced sensor technologies for human detection.
- Implementation of wireless communication and remote control features.
- Publications in peer-reviewed journals and conferences within the last decade.

Screening Process: The initial search yielded many papers, which were then screened based on their abstracts and keywords. Full-text reviews were conducted for papers that met the inclusion criteria.

Audio-based detection

In[5], the authors proposed a microphone array-based sound localisation method for mobile robots. The microphone array employed in the system comprises four microphones positioned in a particular order on the vertices of a regular tetrahedron, since sound has an omnidirectional quality. This study creates two distinct localisation methods based on the time delay of arrival. In a noisy environment, the sound source emits sound. Using a microphone array unit, the sound signal, which consists of the target's sound and noise, is transformed into an electric signal. The localisation unit can process the target's electric signal by amplifying and denoising it in the pretreatment unit. The system records the arrival times of four channels in the localisation unit. The data reached the PC using serial communication of a single-chip for position calculation. In[6], the authors described human search equipment that resembled a robotic car and acted as a backup system to save lives in the event of a catastrophe. Generally, the temperature sensor measures the thermal image of the human body. The robot was equipped with sensors to track people; if an object were found, the readings would change, indicating the presence of a person underneath. The device consisted of motors and an Arduino microcontroller. The module displayed the emission, humidity, and temperature quantities for the individual and the surrounding area using the gas, temperature, and humidity sensors. The system used a Bluetooth module in the device, allowing it to connect to a mobile phone via a mobile application and monitor all movements, including forward, backward, left, and right, with pinpoint accuracy. The Adafruit website was used to collect data directly from the sensors in the module.

In[7], the authors suggested a method for employing an autonomous robot to find human survivors in destroyed areas. The system used a passive infrared sensor (PIR) to identify living people and a camera to capture images of the scene. A feed-forward neural network trained to find a human body or a portion of one in an obstructed environment was given the images. A PC was used to control the vehicle. Three different types of sensors were included in the system: a PIR sensor that used body radiation to identify people in the vicinity, an Infrared (IR) range sensor to identify obstacles in the robot's path, and an image sensor, which was a conventional web

camera, to acquire still photos following instructions from the control program. The Global Positioning System (GPS) was also installed on it. A wireless communication unit provided a reliable communication link with the rescuers. In[8], the authors introduced a portable microwave system to search and rescue people buried in the rubble of a building that had collapsed during an earthquake or other disaster. The system used a continuous-wave X-band radar that could detect heart rate and respiration changes. The data was obtained from the backscattered electromagnetic field using the Independent Component Analysis algorithm (ICA) to separate mixed data. The proposed rescue system also used a remote elaboration unit for data processing. The output signals were given as input to a remote elaboration system utilising a low-frequency wireless channel.

In[9], the authors introduced a system that used an electromagnetic microwave signal for motion detection and could detect the motion of moving objects. This system determined whether or not the object was moving. The working premise of the motion-detecting system was the wave's Doppler frequency change. If an object was moving and present below the earth, the motion detection system used a microwave Doppler radar sensor to detect these waves reflected from the object. Once the motion was identified, it was possible to determine whether or not the item was a human. Also, the device used a microwave test bench to pick up the subject's cardiac and breathing signals. The system consisted of a microwave-frequency electromagnetic signal, a PIC microcontroller, a buzzer, and a display (LCD). The microwave, the LCD, and the buzzer were all connected to the PIC microcontroller. The LCD showed"alive,"and the buzzer turned on when the system detected the motion of moving items. Otherwise, the LCD showed"No Response,"and the buzzer remained off. In[10], the authors designed a rescue robot whose primary job was to locate the robot using a Bluetooth GPS receiver placed on the robot and to find survivors using real-time video transmission. A video camera was connected wirelessly to a laptop for real-time video transmission. The GPS was used for determining the location data for the PC, which contained the address of the area where help was needed. The Microcontroller was used to drive the robot according to the real-time video feedback received on the PC. The Arduino processor was connected to the motor driver responsible for the robot's movement.

In[11], the authors constructed a robotic vehicle prototype controlled by a smartphone using Bluetooth communication. The controller of the robot was an Arduino Uno R3 board. A mobile application was developed to provide an interface between the robot and mobile devices, enabling robot control. The robotic arm also used four servos to facilitate movement. In[12], the authors introduced a rescue robot that used a living human detecting sensor consisting of two components: a"live body sensor"and an"amplifier circuit."The sensor detected the passive Infrared Radiation (PIR) released by living things, and if the movement of the living thing brought on a change, a difference in output was produced. The signal was amplified by a differential amplifier and fed to a PIC microcontroller, which was used to operate the robot's motor and connect it to the robot model.

In[13], the authors introduced a rescue robot that used a PIC-16F877A microcontroller. It could independently navigate through debris and use a temperature sensor to verify the air's temperature. The real-time locating system was used to automatically locate and identify objects or people who were buried in building rubble. The Global System for Mobile Communication Module (GSM) was used to send the measured temperature to the user via Short Message Services (SMS). The robot was also equipped with a camera, LCD, and LED. The wireless camera and PC-Tuner Card recorded the visual and audio signals, where the PC controlled the robot's movement. In[14], the authors proposed a live human detection robot based on Arduino as a controller. It used an ultrasonic sensor to detect barriers and divert its path. It used a PIR that emitted infrared radiation to identify people. When a human target was found, the system started a buzzer and emitted a light alert, which could aid in quickly localising the victim's location. The Arduino received signals from PIR, and according to these signals, the Microcontroller controlled the robot's motors.

Thermal and sensor-based detection

In[15], the authors suggested a system that could detect alive human beings using a set of sensors that indicate the presence of a live human: PIR, temperature, vibration, IR, and ultrasonic detectors. The system used a PIC microcontroller to transmit and analyse human body conditions. Infrared sensors were used to detect a live person. Once the person was located, an audio and visual alarm was quickly sent to the authorities so they could reach the person as soon as possible—the temperature sensor detected and compared the victim's temperature to the previously specified reference temperature. Then, if the measured temperature was higher than the reference temperature, an alert sounded, indicating that the detected body was alive. A wireless camera displayed the path taken by the robot on a laptop. The system used an ATMEGA16 microcontroller to send live human data to a remote-controlled location through the Bluetooth module HC-05.

In[16], the authors designed an autonomous human detection robot to save precious lives. They used PIR sensors to find living people, an AVR Microcontroller, Radio Frequency (RF) modules, temperature and gas leakage sensors, and a camera. The ability to identify motion using PIR sensors was always utilised to determine whether a person had entered or exited the sensor's field of view. When the PIR sensor detected a signal, the system used RF transceivers to communicate with the rescue crew. PIR sensor signals were sent to the Microcontroller, digitised, and sent to the Zigbee network. The RF module transmitted and received data between the robot and the control unit PC.

In[17], the authors suggested a mobile rescue robot to move across an earthquake area, locate living individuals and injured people, and assist rescue operations. The system consisted of a mobile rescue robot and a PC control module. The robot consisted of a sensor unit, a PIC microcontroller, a camera, a motor driver, and a transmission unit. The sensor devices send data to the Microcontroller while monitoring current values. This information was transmitted from the Microcontroller to the PC through a 2.4 GHz RF module. The rescue crew could monitor the readings in real-time through the PC. The camera was used to identify activity and view a live video of the disaster's crucial situation.

In[18], the authors suggested a mobile rescue robot with a PIR sensor that could detect living people in unreachable areas. A joystick controls the semi-autonomous robot, communicating with the control station via RF technology. Ultrasonic sensors were used to detect obstacles in the robot's course of travel, and gas sensors were used to find gas leaks inside buildings. The Microcontroller was programmed using the programming language C and ARDUINO IDE. IP camera integration also allowed for the observation and analysis of circumstances that aided in accurately detecting human activity. The IP camera in the robot is used to observe the situation from a dedicated computer or mobile terminal to help the rescue team find survivors in the disaster-affected area.

In[19], the authors suggested a mobile autonomous robot that moved in an earthquake location and aided in searching for the living and rescue efforts. In this IoT-based project, a live human detector robot could be used for earthquake rescue operations. A PIR sensor was used in conjunction with a temperature sensor to detect the presence of living individuals in a scene, and a PIC microcontroller was used to read the outputs of these sensors. It uses Bluetooth communication to send the data to a PC to be shown on its interface, which has evolved from the convergence of wireless technologies.

In[20], the authors suggested a human detection robot for disaster management, where an Arduino was used to monitor a PIR sensor's output. When the PIR sensor detected a human, the Arduino turned on an LED and a buzzer. False alerts from the PIR sensor could be monitored by the Arduino and taken into account by the operation team. An LCD was used to show the team a human detection notification on the computer. Moreover, a technology consisting of Bluetooth or Wi-Fi can be added for wireless detection robots.

In[21], the authors designed a microcontroller-based human detection robot with a PIR sensor to detect people and an Arduino Uno to control the motors. After receiving the PIR sensor signal, the motors were driven according to the sensor inputs. A mobile application was used to show the detection result, where a Wi-Fi module transmitted the output data from the Arduino.

In[22], the authors suggested a radio-controlled robot that could locate living people in inaccessible areas. A Raspberry Pi was used as the system's controller, and its camera was used to take images of the surroundings using Python libraries. The project collected live image samples and uploaded them to a network to show a live feed on an Android device. These pictures could be used to find people. PIR sensors and 10.525 GHz microwave radar sensors for human motion detection were used to find people trapped under rubble; the robot sent the GPS coordinates of the detected human to the Android device.

Visual and camera-based detection

In[23], the authors proposed an emergency-use, multi-functional sensor-based search and rescue robot system. The system primarily consisted of a camera, an Arduino Mega board, and a Raspberry Pi. The multi-purpose search and rescue robot system could assist in the search and depict the wrecked region from a distance. The system could find metal, look for people in the destroyed area, and transfer the data it had gathered to a web server. The website displayed real-time video streams on the server side. Rescuers could remotely and instantly manage the system, depending on the video streaming. In[24], an autonomous robot for rescue operations was described. The suggested device used an ultrasonic sensor to identify living people and a camera to record a video of the scene as needed. Other sensors, such as temperature, fire, and metal detectors, further detected the presence of bombs during rescue operations. After the ultrasonic sensor detected evidence of a living human, the camera displayed a live scene. After that, the video was shown on the screen. In[25], the authors designed an autonomous robot that could go around an earthquake zone and assist in discovering and recognising live individuals. The system was built using infrared sensors that could recognise human body heat. The autonomous robot could detect each nearby live human-generated thermal radiation with its RIP sensor. The robot then initiated the rescue operation by sending the responsible authorities an audio alarm and its location as soon as it discovered a live human. The robot was also equipped with a stepper motor and a geared DC motor to improve turning precision. The prototype of the rescue robot was made up of various hardware parts, such as embedded microcontrollers, ZigBee transmitters and receivers, RF transceivers, PIR sensors, motor drivers, motion detection, and control PC interfaces.

In[26], the authors proposed a rescue robot that the user can remotely operate, observe, and position. The rescue robot's radar-based technology and GPS capability make it a powerful and capable machine. The user may find the robot and control its direction using this system's live video streaming feature. Moreover, a joystick, an XBee receiver, and other motor types managed by microcontrollers like the Arduino Uno and Arduino Mega combined can all be used to drive the robot. The installed joystick allows the user to send commands. Following that, the user-inserted commands for the rescue robot are sent via the joystick to the Microcontroller for signal mapping and sending. Moreover, the robot has an ultrasonic sensor that measures the desired distance. For the rescue robot to perform its tasks, it is also necessary to assess and estimate the state of the surroundings. On the other hand, the receiver section manages the several servos that control the robot arm's motion. Also, this robot had a mic to send and receive to its ultrasonic microphone.

In[27], the authors proposed a rescue robot with ultrasonic sensors, temperature sensors, and motion detectors to help it avoid obstructions. NodeMCU, IoT, and Android are used to control the rescue robot. The rescue robot is outfitted with communication devices to make rescue mission procedures easier. The robot provides communication between the disaster zones and the volunteer rescuers. Also, the device has sensors implanted that can be used with the application to operate and identify survivors or impediments remotely. In[28], the authors described the implementation of a human-detecting robot that can be used to find people during earthquakes, fire attacks, floods, and other natural and artificial disasters. It's a robot that aids in human detection and can alert rescuers in safer areas to the GPS location of those who need help. The robot also possesses a laser that serves as a rope during floods and as water during fires, and it shoots a straight ray at the location of the human. With a wireless camera and the Viola and Jones method, MATLAB is used to recognise people. When lighting circumstances prevent the camera from functioning correctly, a PIR sensor is utilised as a backup.

In[29], the authors presented a robot that uses a CO2 sensor to monitor a particular area as CO2 levels rise due to exhaled breath. In order to detect the presence of a person nearby, the sensor activates a buzzer when the CO2 level in the area hits a predetermined threshold amount. With the help of a camera and a nearby lamp, the robot can also monitor audio and video. The robot's movement is controlled by a remote, and the person in charge processes the real-time audio and video relayed from the robot to an LCD or TV via RF to draw the proper conclusions about the victim's whereabouts.

In[30], the authors proposed a human-detection robot. It consists of two parts: a transmitter and a receiver. An Arduino Uno controller controls the transmitter side. It receives input from a PIR sensor and an obstacle sensor. The L293D motor drive module, to which a DC motor is linked, and the RF transmitter are the outputs. This system includes wireless communication modules to make it easier to send data. A DC motor can move the robot forward, backward, left, and right. The signals provided by the obstacle sensors are used to determine the direction of movement. An obstacle sensor employs an infrared signal to determine whether there are any objects in its path. If any sensors detect an obstruction, the direction changes to one where there is none. This causes the robot to move automatically without being controlled by an outside source. Another Arduino Uno is used on the receiver side. An RF receiver and a buzzer, respectively, serve as its input and output. The RF receiver alerts the Arduino when it receives the signal from the transmitter. The buzzer then receives a signal from the Arduino, which causes it to start beeping continuously, alerting the user to the presence of a person.

In[31], the authors introduced an autonomous robotic search and rescue system to help find survivors in rubble. This system includes wireless communication modules to make sending data, audio, and video easier. These robots also have sensors and cameras to aid the search process. However, tracking them down was difficult once they got out of the controllers' line of sight.

In[32], an autonomous mobile rescue robot in disaster zones was introduced. The robot consisted of a transmitter and a receiver. The transmitter consisted of a microcontroller with particular sensors, such as a PIR sensor and a raindrop sensor for checking the weather. The receiver unit was an Android phone that received alert notifications when an IR sensor detected human movements, and the buzzer sounded.

In[33], the authors introduced a snake robot that could be used to support the activities of the rescue department. The mobile app shows that the snake robot could have a camera and a gripper in its initial connection, allowing it to search and determine whether people were trapped inside the rubble. If necessary, the gripper on this robot would allow it to pass a hose with air and water to the objective via cracks in the debris. The robot was constructed of several linkages and joints, each separately controlled by servo motors. Bluetooth-enabled wireless control or operation of the robot. They had an Arduino and wireless communication in this robot.

In[34], the authors introduced a snake robot that can move in various environments. This study described a design for a robotic snake that could measure the air temperature and determine whether human life was possible. The robotic snake was of moderate size and equipped with sensors like temperature and ultrasonic ones for simple locomotion. The robotic snake was employed in various applications, including surveillance, monitoring rescue efforts, inspection of confined spaces inaccessible to humans and traditional machines, and determining the temperature of a compact area.

In[35], the authors introduced a robot that could be used to rescue mining personnel who were trapped due to a landslide or other tragedy that occurred in mine sites. Arduino Uno was used as a robot controller. The robot could be moved, its work arm could be used to pick up or hold dangerous mining personnel using a smartphone, and the"Bluetooth RC Controller"Android software could be connected through Bluetooth. In addition, it had an Arduino and wireless communication in this robot, a DC motor, a Bluetooth module, a motor driver, and a battery, which were required.

In[36], the authors created a robotic car by combining open-source hardware, a Bluetooth module, and cutting-edge sensors. They also created Android software that ran on a smartphone and used Bluetooth to monitor and control the robot's performance and access sensor data. The auto obstacle identification and prevention algorithm was programmed into the vehicle and was based on an ultrasonic range sensor. The system also monitored temperature and humidity. The information for these parameters was shown on the Android application being used remotely. A database had been created, which was then examined.

Communication and control systems

In[37], the authors used Bluetooth technology and sensors to create a robotic automobile for indoor environments. They had set up communication between the robot and the smart device through Wireless communication. They had seen the creatures using the phone's camera. An ultrasonic range sensor was kept from clashing with the obstructions in the opposite direction. There is a video cam and a mobile app. In[38], the authors created a robotic arm that was Raspberry Pi-controlled. The primary objective of this robot was to give the robot arm a human arm feature. The Raspberry Pi-based system provided arm movement. The user controlled the robot arm in the desired direction using the Android application over a Wi-Fi connection through wireless communication. This work used an Android application for real-time obstacle identification and avoidance using a remote-controlled, autonomous robotic automobile.

In[39], the authors developed a robotic car that consisted of an Arduino Uno, an H-06 Bluetooth module, an Arduino motor shield, a DC motor, an ultrasonic sensor (HCSR04), a PIR sensor, a buzzer, Wireless communication, a battery, and a microphone. The data was transmitted through Bluetooth from the Android app to the Arduino Uno. Incoming signals were managed by the Arduino Uno, which also instructed which signals should be sent to the motor driver. As a result, based on the inputs entered, the robot moved in a specific order. The user's intelligent device could control the robot's fundamental movements, including back-and-forth movement, rotation to the right and left, and stop motion. The operator could also put the robot in automatic mode and let it go in its direction. The robotic vehicle could detect whether the objects in its path were people or other objects. When an impediment was present, an LED illuminated, the buzzer sounded a warning, and the

shortest distance that could be avoided was determined before moving forward. If the robot faced an inanimate object as an impediment, it calculated the shortest distance it could avoid and moved in that direction. In[40], the authors presented a coal mine detection and rescue robot that could be used to locate and rescue miners trapped in an underground coal mine after a gas explosion in the event of emergencies brought on by natural calamities like explosions. They had created a prototype of a fully autonomous robot that could be used to detect the presence of hazardous gases inside mines for mine rescue operations. Mechanical, electrical, computer, control, communication, sensors, etc., make up its composition. The paper presented the implementation of an embedded control system based on a microprocessor. The embedded control system could control the robot's motion, gather environmental data, and communicate with the remote control system, among other duties.

In[41], the authors introduced a study that describes an Android smartphone and Arduino microcontroller obstacle-avoidance robot. A microcontroller, a video camera, a GPS module, a GSM module, an audio system, and a communication system were typically included as microphones in surveillance robots. Modern smartphones have met all of the requirements for the hardware mentioned above. Using a smartphone significantly reduced the cost of creating a standard surveillance and rescue robot. The robot could be remotely controlled via the Internet via a laptop or tablet. The smartphone camera was used to capture the live video feed. The user received information from the smartphone's sensors, such as geographic location and acceleration. In[42], the authors discussed designing a rescue robot controlled by Arduino that could find people stuck in water or land areas damaged by natural disasters and bring supplies like food and medicine to them. PIR and ultrasonic sensors were used in the proposed system to identify the presence of living things. Moreover, a gas sensor was built to help rescue efforts by detecting the presence of toxic chemicals in the region. A moving mechanism fitted with a float sensor was included to enable the robot to transition from land to water. The system also included a live streaming capability to evaluate the condition of the target living being and a control system for robot movement in all directions via an Android application.

In[43], the topic of this work was the creation of a four-wheeled and Bluetooth-operated search and rescue robot. An Arduino UNO, a motor control shield, a Bluetooth shield, and a mobile app controlled the robot. It was intended to prevent collisions between the robot and an object by utilising a proximity sensor. In[44], the authors introduced the robot"Scouting Owl,"which enabled rescuers to locate and contact people in dangerous situations that were not yet suited for physical rescue. They had created a robot based on a Raspberry Pi 4 to carry out the crucial rescue task. The robot featured a dependable half-duplex connection between the robot and the control device, which fetched and displayed real-time sensor data and an IR camera feed on the display device, and controlled the robot remotely via an Android app. Among the sensors and actuators were a flame sensor, a water level sensor, an infrared camera, an ultrasonic sensor, an electromagnetic pulse sensor, a temperature sensor, a DIY radiation sensor, a speaker, and a microphone.

In[45], the authors proposed a movable robotic system to find people in unmanned geographic constructions. The system had Arduino, which consisted of equipment and monitoring sensors for evaluating the state of the human body and sending data to a server. These mobile robots used localisation tracking for wireless sensor network communication. A camera unit, a PIR sensor, a PC as a processor, high-frequency-based motion and existence detection, and a GPS module for latitude and longitude detection were all part of the sensing system on the mobile robot. In[46], the authors introduced a remote-controlled automobile that could travel in all four directions. The war field robot consisted of an Arduino board as a controller board. The ZigBee transceiver module wirelessly accepted control data from the Remote Control Unit (RCU). The Microcontroller controlled the robotic rover motor. The motor was linked to the Microcontroller through relay drivers. The system detected pulses and temperature fluctuations. If rays were identified, a high value was given to the Microcontroller. The Microcontroller then read the GPS receiver's latitude and longitude and sent them using the ZigBee transceiver to the control unit. The ZigBee transceiver delivered and received data between the robot and the control unit.

In[47], the authors suggested an autonomous robotic vehicle that travelled through an earthquake-prone area and aided in identifying the living and assisting rescue efforts. PIC16F877A is the Microcontroller used in the proposed method. The system used a detector and PIR sensor to detect the person's body temperature buried beneath the dirt. The PIR provided a signal to the CPU when it detected the physical body's temperature. The data would be collected from ZigBee and sent to the PC via a voltage conversion circuit and a PC interface unit. The signal was transmitted via ZigBee to the interface unit, followed by an alarm sounding and a visual display to notify the appropriate officials. In[48], the authors suggested utilising an ultrasonic sensor to identify a live human, where a mobile robot was created. The robot had a camera to record a live video and deliver it to the rescuer. It had a PIR sensor for person detection and an ultrasonic sensor for obstacle detection. The rescuer's device showed a video of a live individual in a rescue or dangerous location. This could be used to find burglars or terrorists hiding inside the structure. It could also be used to find living people in buildings that had fallen due to natural disasters like earthquakes.

In[49], the authors introduced a system that could be housed in a robotic vehicle to be used in disaster-prone places and was based on a PIC microcontroller and a ZigBee transmitter and receiver. The suggested autonomous robotic vehicle drove in earthquake-prone areas and aided in locating survivors and in rescue operations. A remote control was made for short distances. A camera module was mounted on the robot and used as a web camera. The video feed was delivered to the receiver in the control unit on the PC using this camera. This camera module broadcasted video footage of the routes that the rescue team should follow. When the system detected human movement, the GPS receiver obtained the longitude and latitude values. Utilising a ZigBee transmitter, this data was sent to the control unit and shown on an LCD. This part (the receiver unit) also controlled the robot's motion using the PC. In[50], the authors suggested developing a surveillance robot car driven by an Android smartphone and controlled by an Arduino UNO microcontroller. A smartphone was used to control the robotic car remotely using Bluetooth communication. A built-in camera on the robot and Android smartphones is used as input to control four motors, allowing for real-time video recording from the

robot vehicle. To prevent a collision, the surveillance robot employed an ultrasonic sensor to identify any objects in front of the moving vehicle. If the robot detected any obstacle, it stopped itself.

In[51], the authors suggested a compound robot. The primary intent of this project is to design and bring about a robot prototype using Arduino Uno, Motor Driver L293D, and HC05-Bluetooth module. The goal of this project is to gain knowledge about the Ultrasonic sensor HCSR-04, reconcilable software, and a controlled motor circuit. The compound robot could move in four directions using an Android app. The robot used an ultrasonic sensor to detect the distance between the robot and the obstacles, which was displayed all the time on the mobile app. In[52], the authors introduced a live human detection robot that could be used in rescue operations. It used an AVR microcontroller, a PIR sensor, and other components. The robot used a PIR sensor to measure human body temperature. When it detected an alive person, an alert or indication signal was sent, and a message was delivered through SMS utilising GSM technology to enable rescue operations.

In[53], the authors suggested a human-detection robot. The robot was equipped with a camera to transmit live video of the disaster location to the rescue team on any Android device or laptop. The human detection module also utilised infrared LED lighting to illuminate the robot's surroundings under low or no light conditions. The robot's human detection module primarily consisted of a PIR sensor that an Arduino managed. The PIR sensor could detect and continue to signal, but it ceased when the motion was no longer there. Additionally, the person-detecting module featured infrared lights. In[54], the authors introduced a system of drones for earthquake rescue operations. A PIR, an Atmega8A microprocessor, and a buzzer served as the main components of the person detection system. The PIR was used to detect the presence of a living human body. Control of the quadcopter was accomplished by RF technology. Once a living person had been found among the rubble, the Microcontroller immediately received a signal, digitised the signal, and turned on a buzzer. A radio receiver attached to the flight controller received the orders issued by the transmitter, which utilised radio signals to operate the quadcopter wirelessly. The rescue operators of the impacted locations could receive an alarm message from the ATMEGA8A microcontroller and then provide the afflicted victims with the necessary rescue.

In[55], the authors suggested the design of an intelligent rescue robot with image processing. The system used an Arduino Uno and a Raspberry Pi. The robot's movements were managed automatically or manually using the Microcontroller Arduino Uno. The Raspberry Pi, which had an image processing system, was used to review the camera's recorded image. When impediments were close to the robot, an ultrasonic sensor was activated. It signalled the Arduino Uno board, determining the robot's movement or activity. In order to change the configuration of robots from automated to manual motion, they employed visual processing to recognise people. A camera was installed on the robot, which was used to take pictures and send them to the controller for image analysis.

In[56], authors have presented a data-driven approach for detecting human skin in cluttered search and rescue environments (DDHS), a crucial task for locating and identifying survivors after disasters like earthquakes. The authors created a novel dataset of skin and non-skin patches captured in simulated search and rescue scenarios, which included challenging factors like varying lighting, occlusions, and cluttered backgrounds. Using this dataset, they trained a convolutional neural network (CNN) model to classify image patches as skin or non-skin, designing the model to be robust to the challenges present in search and rescue environments. The proposed skin detection model demonstrated superior performance compared to traditional colour-based and machine learning-based approaches when evaluated on the authors' dataset and other skin detection benchmarks, and its effectiveness was further validated in a simulated search and rescue setting. This work contributes to advancing computer vision and robotics technologies for disaster response and search and rescue applications, where rapidly identifying survivors is critical for saving lives. In[57], the authors presented a hybrid human detection (HHD) technique that can quickly locate isolated victims during indoor disasters, even when they are obscured by smoke from fires. The HHD method achieves a higher human detection rate by merging the one-stage object detection algorithms YOLO and RetinaNet. It can identify victim locations using existing building CCTV cameras and overlaying images with a smoke filter without requiring expensive specialised equipment. Experiments showed that the HHD technique has faster detection speed and higher accuracy than prior methods. It is well-suited for the critical early stages of an indoor disaster response when rapidly finding and rescuing trapped victims is crucial to prevent secondary injuries or fatalities.

According to the technologies used in each related work and as shown in Fig. 1, Fig. 2, and Table 1, the studies can be summarized as follows:

- Arduino has been used as a microcontroller in 25 papers; PIC has been used in 10 papers; five papers used Raspberry Pi, while two papers used a combination of Raspberry Pi and Arduino.
- 46 papers involved wireless communications, 6 of which used Wi-Fi, and 17 used Bluetooth
- A microphone was used in 7 papers.
- Twenty-seven papers used a video camera, and one used a mobile camera.
- No papers used a thermal camera to detect the presence of live people, and 25 papers used a PIR sensor to detect live people.
- Twenty-one papers used mobile applications to control the robot.
- Twenty-two papers used computer devices to control the robot.

We therefore propose a reliable prototype of SERR compared with the existing research in the literature.

Novelty and contributions of the SERR research work

The proposed SERR research work has many novel contributions that are different from the articles discussed above:

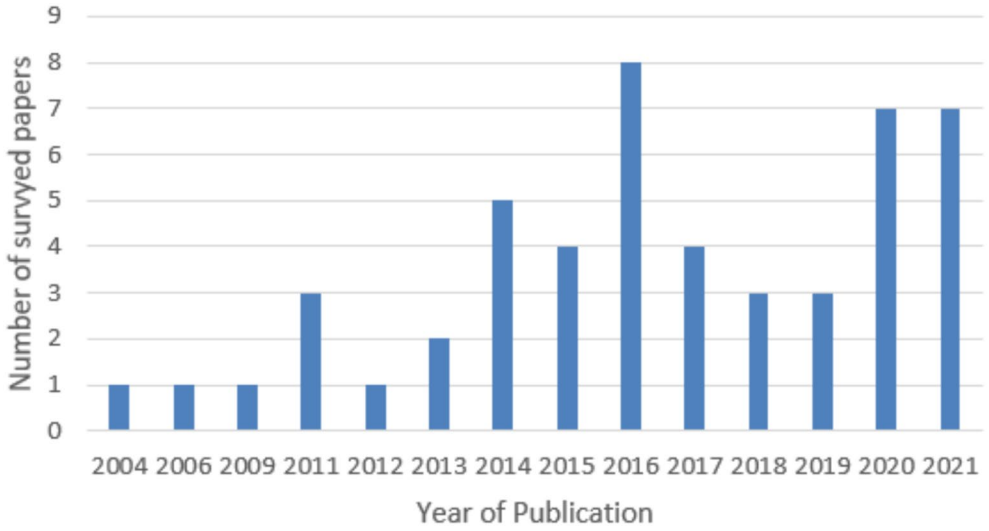


Fig. 1. Number of surveyed papers published in each year of rescue robot systems.

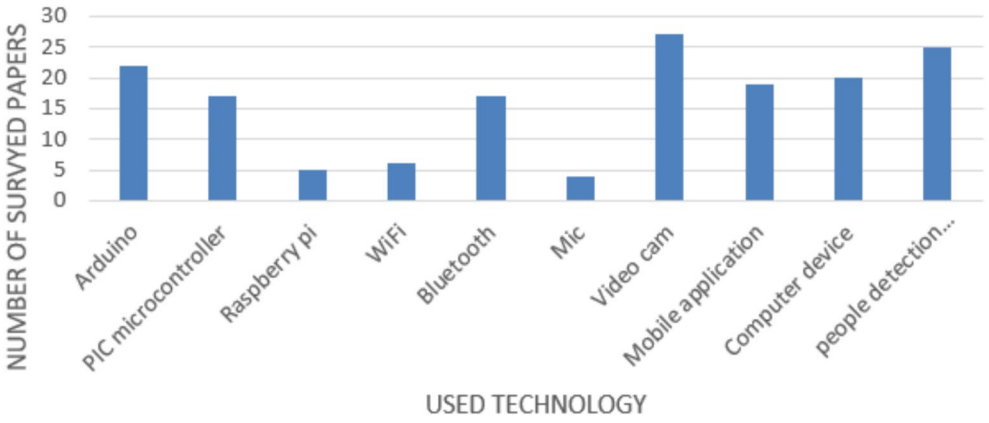


Fig. 2. Technologies used in surveyed papers.

This paper differs from previous studies, concentrating on a single sensor by combining thermal imaging and camera systems. The purpose is to enhance the precision and dependability of identifying trapped individuals. Using many sensors improves the robot's capacity to detect and recognise humans in different situations, including those with limited visibility or when debris is present.

This research focuses on developing reliable wireless communication systems that provide immediate contact between the robot and the rescue team rather than simply investigating remote control capabilities, as earlier studies have done. Live audio and video feeds greatly enhance situational awareness and collaboration in rescue operations.

This article emphasises the practical application of the robot by providing specific design considerations and implementation processes for creating a workable prototype instead of discussing theoretical concepts. Focusing on practical implementation ensures that the findings can be immediately applied in real-life rescue situations, thus eliminating the gap between research and practical use.

The proposed design prioritises detecting humans rather than navigating obstacles, which is of utmost importance in earthquake rescue situations. While other research may prioritise barrier avoidance, this paper highlights the significance of locating and rescuing humans, making it especially applicable for disaster response applications.

Although the reviewed studies present mixed views on earthquake rescue robots, the Smart Earthquake Rescue Robot (SERR) has been proposed and developed, and offers several unique features and improvements:

Integrating several advanced technologies: SERR includes live video streaming, heat detection using Grid-Eye sensors, and wireless remote control in one compact system.

Improved Human Recognition Capabilities: Using both visual and thermal data, SERR can more accurately locate survivors, including unconscious people.

Improved Communication: SERR has a built-in speaker and microphone. Allows for two-way communication between the rescuer and survivor.

References	Year	Arduino	PIC	Rasp.pi	Wireless comm	Mic	Video cam	Mobile app	Computer	People detection
Audio Based Detection Methods										
[5]	2009					√			√	
[6]	2021	√			√			√		
[7]	2014				√		√		√	√
[8]	2011				√					
[9]	2017		√							√
[10]	2013	√			√		√		√	
[11]	2016	√			√			√		
[12]	2016		√							√
[13]	2014		√		√		√		√	√
[14]	2018	√								√
Thermal and Sensor Based Detection Methods										
[15]	2017		√		√		√		√	√
[16]	2018				√		√		√	√
[17]	2014		√		√		√		√	√
[18]	2016	√			√		√		√	√
[19]	2020		√		√				√	√
[20]	2021	√			√				√	√
[21]	2021	√			√			√		√
[22]	2021			√	√		√	√		√
Visual and Camera Based Detection Methods										
[23]	2020	√		√	√		√		√	
[24]	2011				√		√		√	√
[25]	2013		√		√		√		√	√
[26]	2015	√			√	√	√	√		
[27]	2020				√			√		√
[28]	2017				√		√		√	√
[29]	2011				√		√			√
[30]	2015	√			√					√
[31]	2009				√		√			√
[32]	2020		√		√	√		√		√
[33]	2021	√			√		√	√		
[34]	2016	√			√		√		√	
[35]	2021	√			√			√		
[36]	2016				√			√		
Communication and Control Systems Based Detection Methods										
[37]	2014	√			√		√	√		
[38]	2015			√	√			√		
[39]	2019	√			√	√		√		√
[40]	2012								√	
[41]	2016	√			√	√	√	√	√	√
[42]	2019	√			√		√	√		√
[43]	2018	√			√			√		
[44]	2022			√		√		√		√
[45]	2017	√			√	√	√	√	√	√
[46]	2020	√			√		√		√	√
[47]	2015		√		√				√	√
[48]	2016	√			√		√			√
[49]	2019		√		√		√		√	√
[50]	2020	√			√		√	√		
[51]	2016	√			√			√		
[52]	2014				√					√
[53]	2016	√					√		√	√
[54]	2018				√					√
[55]	2020	√		√	√		√	√		√
Continued										

References	Year	Arduino	PIC	Rasp.pi	Wireless comm	Mic	Video cam	Mobile app	Computer	People detection
[56]	2020	√	√	√	√	√	√	√	√	√
[57]	2022	√	√	√	√	√	√	√	√	√
SERR	2024	√			√		√	√	√	√

Table 1. Related work technologies summary.

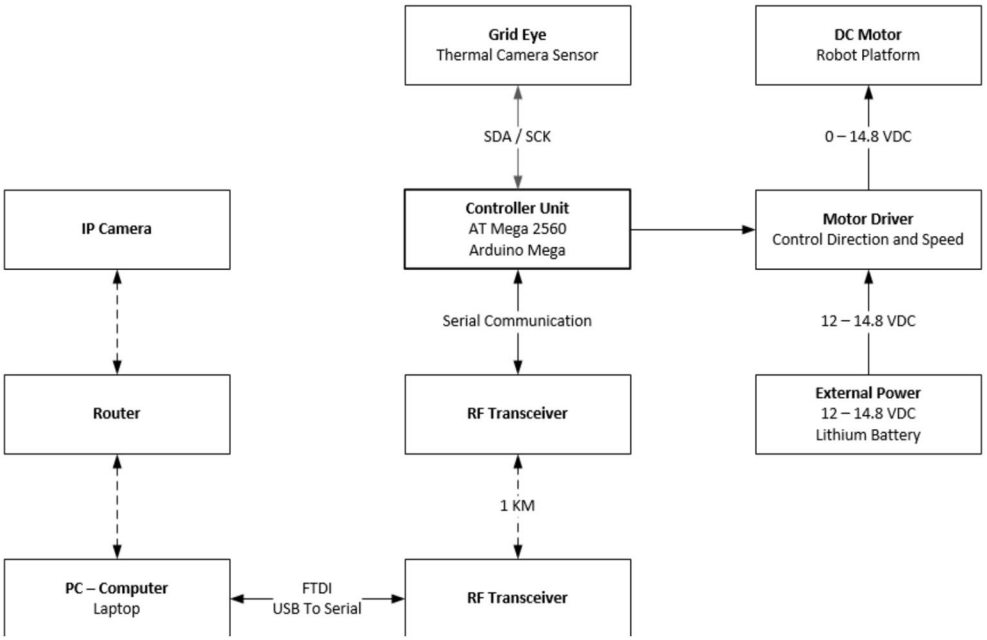


Fig. 3. Block diagram of SERR.

User-friendly interface: Smartphone application for remote control provides rescue teams with an intuitive and easy-to-use operating tool.

Comprehensive Performance Evaluation: Our study includes extensive testing using MATLAB simulations and real-world datasets. It provides a detailed assessment of SERR's potential.

Design of proposed SERR

Mechanical design and mobility features

The proposed SERR consists of a two-way communication system with a Grid eye, an ultrasonic sensor, a camera, an Arduino microcontroller, and an RF module. The SERR chassis uses a reinforced four-wheel drive with shock-absorbing suspension, allowing traversal over uneven debris up to 20 cm in height. The frame is constructed from lightweight aluminum, balancing durability and portability. The battery enables 3 h of continuous operation. Stability is maintained using a low center of gravity, and the wheel treads are designed for traction on loose rubble. Future versions will incorporate articulated tracks for improved performance on unstable terrain. The block diagram of the proposed SERR design is shown in Fig. 3. The system architecture diagram (Fig. 3) provides a comprehensive overview of the SERR's hardware and software integration. Figure 3 illustrates the integration of hardware components, including the Grid-Eye thermal sensor, ultrasonic sensor, camera, Arduino Mega microcontroller, RF module, and user interfaces (PC and smartphone applications), with data flow and communication pathways. It highlights the connections between the Grid-Eye thermal sensor, ultrasonic sensor, camera, Arduino Mega microcontroller, RF module, and user interfaces (PC and smartphone applications). Data flow directions and inter-module communications are indicated, supporting real-time human detection and remote operation. It provides an excellent solution to help rescue operations after an earthquake by searching for people under rubble instead of rescue teams, thereby avoiding exposure to dangers and risks. The SERR is non-complicated and easy to use by the users; it mainly consists of a 4-wheel robot chassis and a motor driver. The power supply of the SERR consists of a rechargeable Lead-Acid battery and a protection fuse.

Sensor selection and integration

A PC application based on an RF module is developed to control the SERR wirelessly. Once the user turns the SERR on and connects the smartphone application to it, a live video stream appears on the smartphone application. The output of Grid Eye appears on the PC application, which detects the presence of living humans

or animals under the rubble. The main objective of the Smart Earthquake Rescue Robot (SERR) is to search for and identify survivors in post-earthquake situations efficiently. Although detecting and avoiding obstacles is important, the SERR robot design prioritized human detection capabilities. In practical use, SERRs must be designed to move over or around obstacles. Instead of stopping when confronted, the design is simplified by focusing on core functionality.

To achieve this goal, the SERR robot combines several functionalities:

Survival Search: The robot can use cameras and thermal sensors to search for human or animal survivors beneath the rubble.

Obstacle-ridden travel: Robots have autonomous navigation capabilities to navigate safely through chaotic environments.

Remote control: A smartphone application allows rescue teams to control robots remotely. This ensures that the robot is directed precisely to the desired point.

Communication and control

Communication: Although not implemented in the current prototype, the design also has provisions for two-way communication between the rescuer and the survivor via a built-in speaker and microphone.

The flowchart of the SERR is shown in Fig. 4. Figure 4 illustrates the flowchart of the SERR operational process, detailing the sequence of user interaction, wireless connection via RF module, live video streaming, and motor control based on user input and obstacle detection. The user should first turn the SERR on and then connect the PC application to the SERR via a wireless RF module connected to the PC with USB. The SERR streams a live video on the mobile application. If the user presses any movement button, the controller moves the SERR motors according to the required direction. The SERR stops when it approaches any obstacle.

The overall process of SERR is shown in Fig. 5. Figure 5 illustrates the overall process of SERR in rescue operations, depicting the workflow from situation evaluation, survivor detection using thermal and visual sensors, to remote control and communication facilitated by rescuers via PC and smartphone applications. The first tasks in rescue operations are to evaluate the situation, locate the victims, and contact them. The SERR searches for survivors trapped in rubble and controls it wirelessly through an RF transceiver by a PC. It has a camera for streaming live video controlled by a smartphone application by rescuer 1 and can communicate by sound through camera speakers and a microphone. Grid Eye enables rescuer 2 to detect the presence of living humans by detecting temperature, as indicated by the colour in the PC application. Then, the user can move the SERR in all directions and stop. Using an ultrasonic sensor, the SERR can stop when it is near any obstacle.

Critical human detecting functionality has been given significant attention in the design of the Smart Earthquake Rescue Robot (SERR). This feature is given more priority than obstacle detection to ensure that the primary objective of rescuing people is accomplished. The present SERR prototype has limited autonomy when navigating obstacles, but that will change in subsequent versions with the integration of better navigation technologies. Human detection through thermal imaging and video systems is still the prototype's primary goal. The design incorporates features that allow the SERR and rescuers to communicate, which enhances the robot's utility in rescue missions. Even though it is not entirely functional in the prototype, this functionality is essential for coordinating rescue attempts and giving real-time updates. The SERR may communicate with the

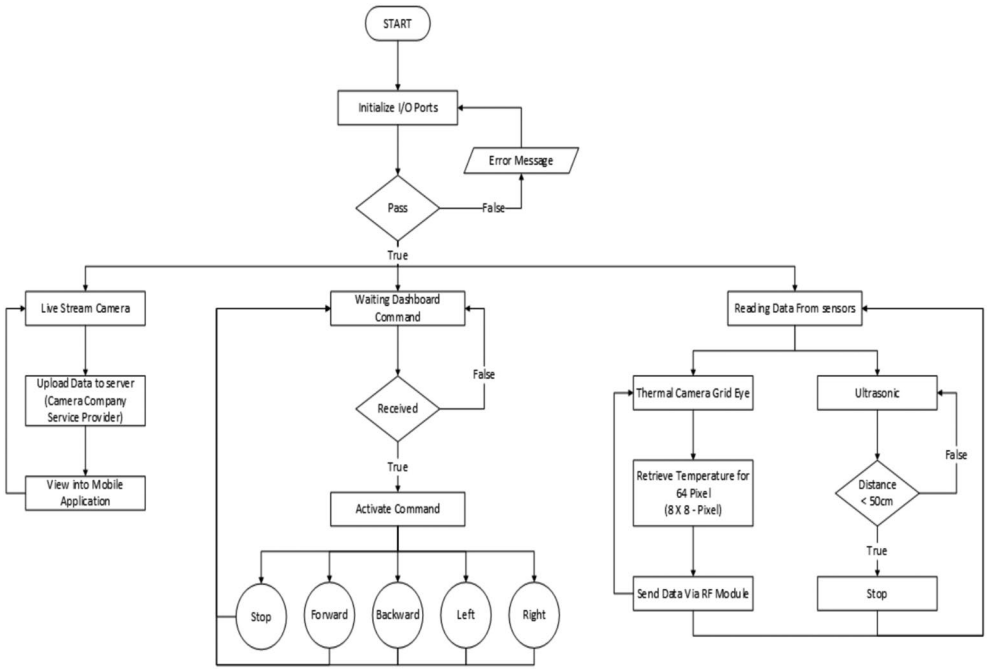


Fig. 4. Flowchart of SERR.

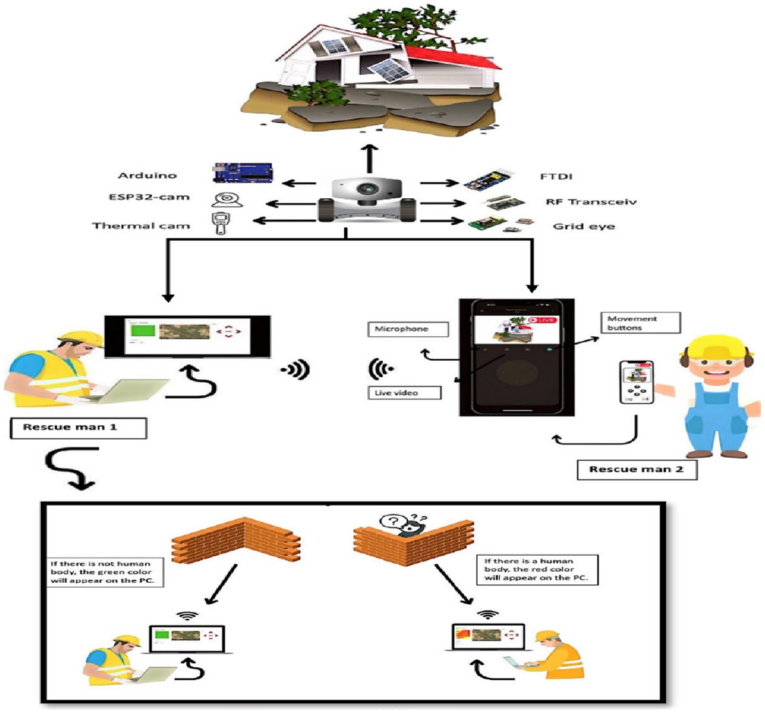


Fig. 5. Overall process of SERR.

rescue team through audio signals sent and received by its integrated speakers and microphones. This capacity enhances the efficiency and effectiveness of rescue operations by allowing rescuers to interact directly with individuals trapped under debris.

Workflow of the SERR system

This study explains the signal-capturing and data-processing operations for the sensors used in the Search and Rescue Robot (SERR) model. The automatic adaptation mechanism in the SERR, based on real-time feedback from sensor data, involves analysing the sensor readings and dynamically adjusting the threshold distance for obstacle detection. We employ cross-correlation and time-of-flight estimation techniques for the ultrasonic sensor to capture and process the received signal, allowing us to calculate the distance to objects. The infrared sensor utilises calibration parameters and statistical methods to capture and process the intensity of infrared radiation, enabling temperature estimation. Finally, the camera sensor captures images or video frames, and it employs convolutional neural networks (CNNs) for object detection and recognition, employing convolution operations and non-maximum suppression techniques. The overview of the system workflow:

Continuous sensor data monitoring

The SERR continuously monitors the sensor data from the ultrasonic sensor in real-time. The sensor data includes the measured distances to objects or obstacles in the SERR's vicinity.

Data analysis and statistical processing

The SERR's software or algorithm processes the sensor data to analyse the distribution and characteristics of the measured distances. Statistical techniques such as mean, standard deviation, or clustering algorithms can be applied to identify patterns and trends in the data.

Dynamic threshold adjustment

Based on the sensor data analysis, the SERR dynamically adjusts the threshold distance for obstacle detection. The adjustment can be made based on factors such as the average distance to detected objects, the distance variation, or the presence of outliers. The aim is to set a threshold distance that optimises obstacle detection while minimising false positives or negatives.

Feedback loop

The SERR continually evaluates the performance of the adjusted threshold distance. It compares the detected obstacles based on the adjusted threshold with the actual environmental obstacles. The feedback loop allows the SERR to assess the effectiveness of the current threshold and make further adjustments if necessary.

Iterative refinement
The automatic adaptation mechanism operates iteratively, continuously updating the threshold distance based on real-time feedback. As the SERR encounters different obstacles and environments, it refines the threshold to improve the accuracy and reliability of obstacle detection.

Hardware modules and their operation in SERR
The SERR integrates key hardware components, including an ultrasonic sensor for obstacle detection (halting at 30 cm), a Grid-Eye thermal sensor for human presence detection via temperature mapping, and a camera for live video streaming, all of which are controllable via a smartphone app. An Arduino Mega microcontroller processes inputs and manages a 4-wheel chassis with shock-absorbing suspension, powered by a rechargeable Lead-Acid battery (3-h operation). An RF module enables wireless control through a PC application. Detailed signal processing techniques (e.g., cross-correlation for ultrasonic data, temperature calibration for Grid-Eye) and hardware specifications are provided in Appendix A.

SERR's obstacle detection and distance-based navigation Algorithm

The SERR algorithm consists of the following variables –

T: Temperature value obtained from the Grid Eye sensor.
D: Distance measured by the ultrasonic sensor.
C: Color representation based on temperature (red, orange/yellow, green).
O: Obstacle detection signal (1 for True, 0 for False).
M: Motor control signal for the SERR's wheels.

The constraints applied to the algorithm are:

threshold_high: Upper-temperature threshold for"Red"color.
threshold_low: Lower temperature threshold for"Green"colour.
threshold_distance: Threshold distance for obstacle detection.

The SERR algorithm manages obstacle detection and navigation by integrating sensor inputs such as temperature (T) from the Grid Eye sensor and distance (D) from the ultrasonic sensor. The algorithm determines a color representation (C) based on temperature thresholds (threshold_high and threshold_low) and sets an obstacle detection signal (O) using the ObstacleDetection function. If an obstacle is detected (O = 1), the MotorControl function halts the SERR (M = 0). Otherwise (O = 0), motor control (M) is adjusted based on the desired distance to navigate forward, backward, or stop. Detailed pseudocode for this process is provided in Table 2.

Implementation and testing of proposed SERR

The SERR prototype includes three main components: the Grid Eye, the ultrasonic sensor, and the RF module. The Grid Eye provides a thermal image of the target closest to a thermal camera and sends the data to the Arduino microcontroller. Then, the data is sent to a computer through the RF module. The computer receives data and draws it using the dashboard. To control the motion of the SERR by the PC, and work on the dashboard using the VISUAL BASIC 2010 version. The first step is to open the port from the Device Manager. Then select

```
TemperatureColor(T):
        C = { "Green" if T < threshold_low,
        "Orange/Yellow" if threshold_low ≤ T ≤ threshold_high,
        "Red" if T > threshold_high }
ObstacleDetection(D):
        O = { 1 if D ≤ threshold_distance,
        0 if D > threshold_distance }
MotorControl(D, O):
        M = { 0 if O is 1,
        DesiredSpeed if D < DesiredDistance,
        -DesiredSpeed if D > DesiredDistance,
        0 if D = DesiredDistance }
```

Table 2. SERR Algorithm.

the port number (com5) and open it to be ready to use, as shown in Fig. 7. The next step is to go to the application and select communication, as shown in Fig. 8. Three sections appear in the window:

1. The Grid Eye section comprises 68 pixels, arranged in an 8 × 8 grid. Each pixel's background colour can be adjusted based on the detected body temperature:
2. A red colour indicates high temperatures.
3. Moderate temperatures are shown as orange or yellow.
4. The green colour represents low temperatures.
5. The feedback Section down the board shows us the output of the grid eye.
6. The control section moves the robot toward the arrow or stops the robot.
7. Straightforward direction.
8. Back direction.
9. Right direction.
10. Left direction.
11. Stop.

Openings were created in the outer shell of the SERR to accommodate the ultrasonic sensor and Grid-Eye module so that they can work adequately, as shown in Fig. 6. Figure 6 illustrates the physical prototype of the SERR, showcasing the 4-wheel chassis, sensor placements (ultrasonic and Grid-Eye), and structural design for navigating debris. Ultrasonic stops the SERR at a distance of 30 cm from objects, as shown in Fig. 7. Figure 7 illustrates the SERR Operation at a Distance of 30 cm from Objects, demonstrating the ultrasonic sensor's ability to halt the robot to prevent collisions. Additionally, the SERR final product aims to incorporate communication facilities to assess and interact with distressed victims, further facilitate their exact location, and provide emergency compartments for water and food.

When there is no near object in front of the SERR, the green colour appears in the Grid Eye section, allowing the SERR to continue moving, as shown in Fig. 8. Figure 8 illustrates the moving case of SERR, showing the Grid-Eye section displaying green color on the PC application, indicating no nearby objects and allowing continued movement. When there is a near object in front of the SERR, the red colour appears in the Grid Eye section of the application and stops the SERR, as shown in Fig. 9. Figure 9 illustrates the stop case of SERR, displaying red color in the Grid-Eye section of the PC application, indicating a nearby object and triggering the robot to stop.

The key metrics assessed were accuracy, precision, recall, F1-score, and runtime. The accuracy of the SERR in detecting the presence of trapped individuals under the rubble was found to be 0.94, indicating a high rate of correct predictions. The precision, which measures the proportion of true positive detections among all positive detections, was 0.90, demonstrating the system's ability to minimise false alarms. The recall, representing the proportion of true positive detections among all actual positive instances, was 0.96, suggesting the SERR's effectiveness in identifying most trapped individuals. The F1-score, the harmonic mean of precision and recall, was 0.92, further validating the system's overall performance.

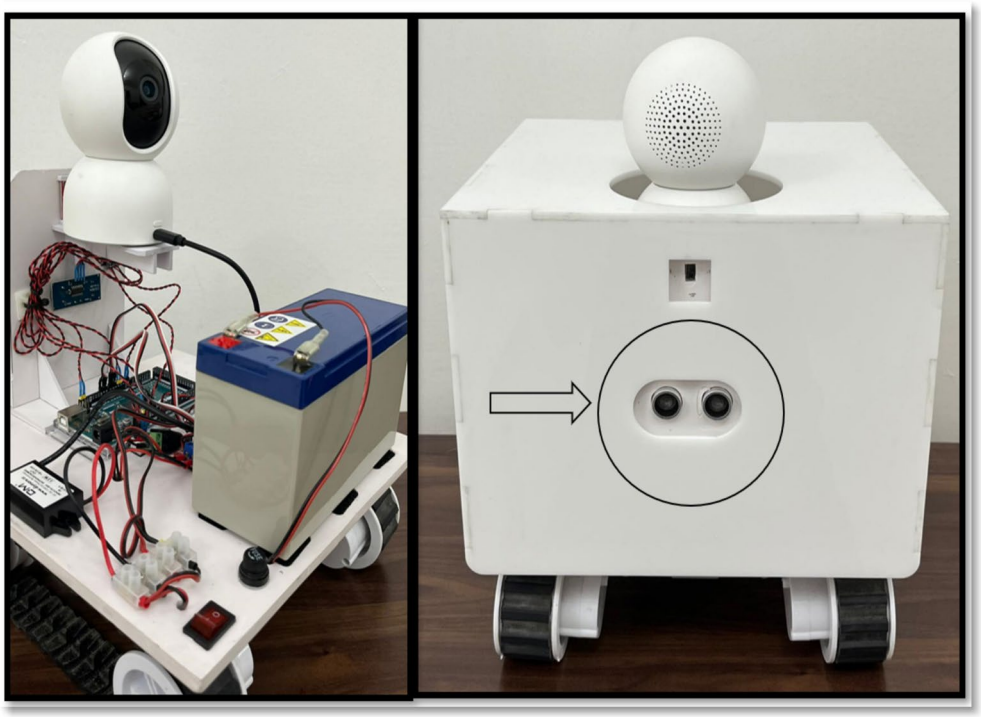


Fig. 6. Prototype of SERR.

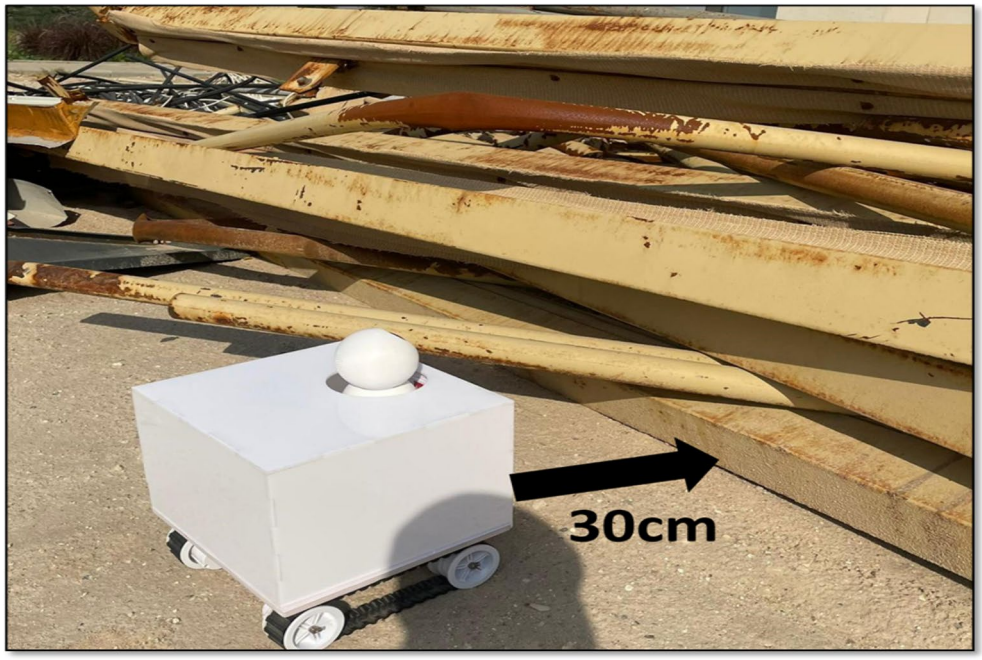


Fig. 7. SERR at a distance of 30 cm from objects.



Fig. 8. Moving case of SERR.

Additionally, the SERR exhibited a runtime of 35 ms, which is crucial for real-time operation in emergency scenarios. This fast response time ensures that the system can provide timely information to rescue teams, enabling them to act swiftly and efficiently in locating and assisting trapped victims. These results indicate that the proposed SERR prototype is a promising solution for earthquake rescue operations, providing a reliable and efficient means of detecting and locating survivors trapped under debris. The combination of high accuracy, precision, recall, and fast runtime makes the SERR a valuable tool for enhancing the effectiveness of search and rescue efforts in the aftermath of natural disasters. Additionally, the SERR final product aims to incorporate



Fig. 9. Stop case of SERR.

communication facilities to assess and interact with distressed victims, further facilitating their exact location, and provide emergency compartments for water and food. However, it is acknowledged that the two-way audio communication feature via built-in speakers and microphones, while designed into the system, is not yet implemented or functional in the current prototype. Future development will focus on activating and testing this capability to enable direct interaction between rescuers and survivors.

Algorithm development (RescueNet)

To enhance the effectiveness of SERR, it integrated a novel machine-learning model designed for real-time data analysis. This model, named RescueNet, is a Convolutional Neural Network (CNN) with Long Short-Term Memory (LSTM) layers tailored explicitly for analysing the data transmitted by the robot, including video feeds and thermal images. The functionality of the algorithm is:

- Survivor Detection: Processes live video and thermal images to detect signs of life accurately.
- Obstacle Recognition: Identifies and categorises obstacles, enhancing the robot's autonomous navigation capabilities.
- Communication Signal Detection: Analyses audio signals to detect and amplify faint sounds from trapped victims.

The RescueNet model is comprised of:

- Data Preprocessing: The input data (video, thermal images, and audio signals) are preprocessed to normalise and enhance features relevant to detection and classification.
- CNN Layers: Extract spatial features from the video and thermal images.

  LSTM Layers: Capture temporal dependencies and patterns in the data, which are crucial for identifying continuous signals from survivors.

- Output: The model outputs detection results in real-time, providing coordinates and confidence levels for identified survivors and obstacles.

The RescueNet architecture consists of three convolutional layers for both video and thermal image streams, each with 32, 64, and 128 filters, respectively (kernel size 3×3, ReLU activation), followed by max pooling (2×2). The output feature maps are flattened and concatenated with extracted audio features (e.g., Mel-frequency cepstral coefficients—MFCCs). The combined feature vector is input to a two-layer LSTM network with 128 units per layer. The output is fed into two fully connected layers (one each for survivor and obstacle detection), each followed by a softmax activation. The model is trained using the Adam optimizer with a learning rate of 0.001, categorical cross-entropy loss, and a batch size of 32 for 50 epochs. Early stopping is applied to prevent overfitting.

The pseudocode of the RescueNet model is given below.

```
Input:
        - Video frames (V)
        - Thermal images (T)
         - Audio signals (A)

Preprocessing:
  For each video frame and thermal image:
     Normalize pixel values
     Resize to input dimensions
  For audio signals:
     Extract relevant features (e.g., MFCCs)

Feature Extraction (CNN):
  For each image (V, T):
      Pass through a series of convolutional layers with ReLU activation
      Apply max pooling after each conv layer
      Flatten output feature maps

Feature Fusion:
  Concatenate flattened video, thermal, and audio features into a single feature vector F.

Temporal Modeling (LSTM):
  Pass a sequence of feature vectors F through LSTM layers
  Output hidden states H

Detection Heads (Fully Connected + Softmax):
  Survivor detection: FC layer + softmax (output: probability, coordinates)
  Obstacle detection: FC layer + softmax (output: probability, coordinates)

Output:
  For each time step, return detection probabilities, locations, and confidence scores for survivors
and obstacles.
```

RescueNet Algorithm and operation

Data Preprocessing: $X=\{X_{video}, X_{thermal}, X_{audio}\}$

Feature Extraction: Convolutional layers $Conv_{k,l}$ with kernel size k and stride l

$$F_{video} = Conv_{3,1}(X_{video})$$

$$F_{thermal} = Conv_{3,1}(X_{thermal})$$

Pooling layers for down-sampling: $P_{video} = MaxPool_{2,2}(F_{video})$

$$P_{thermal} = MaxPool_{2,2}(F_{thermal})$$

Temporal Feature Extraction using LSTM:

Flatten the pooled features and concatenate them with audio features

$$P = Flatten(P_{video}) \oplus Flatten(P_{thremal}) \oplus X_{audio}$$

LSTM layers LSTMm with m units

$$H=LSTM_{128}(P)$$

Output Layer:

Fully connected layer FC_n with n units

$$Y_{survivor} = Softmax(FC_2(H))$$

$$Y_{obstacle} = Softmax(FC_3(H))$$

Coordinates and confidence levels for detected survivors and obstacles

$$C_{survivor}, O_{bstacle} = Coordinates(Y_{survivor}), Coordinates(Y_{obstacle})$$

$$Conf_{survivor}, Conf_{obstacle} = Confidence(Y_{survivor}), Confidence(Y_{obstacle})$$

Figure 10 illustrates the architecture of RescueNet, a multimodal CNN-LSTM model for SERR, showing input data streams (video, thermal images, audio) processed through three CNN layers (32, 64, and 128 filters, with a max pooling of 2×2) for spatial feature extraction. Features are concatenated post-pooling into a single vector, which is then fed into a two-layer LSTM (128 units each) for temporal modelling over 10 frames. The output is provided via two fully connected heads (survivor and obstacle detection) with softmax activation, accompanied by confidence scores (threshold 0.7). Data flow and integration points are highlighted, demonstrating RescueNet's capability for real-time survivor detection in disaster scenarios.

Data Preprocessing: The input data X, comprising video, thermal images, and audio signals, is preprocessed to normalise and enhance relevant features. This step ensures that the data fed into the model is consistent and highlights essential features for detection and classification.

Feature Extraction Using CNN: Convolutional Neural Networks (CNNs) extract spatial features from the video and thermal image data. The convolutional layers apply filters to detect features such as edges, textures, and patterns.

Pooling layers down-sample the feature maps, reducing their dimensionality while preserving important information. This step helps reduce the computational complexity and retain the most relevant features.

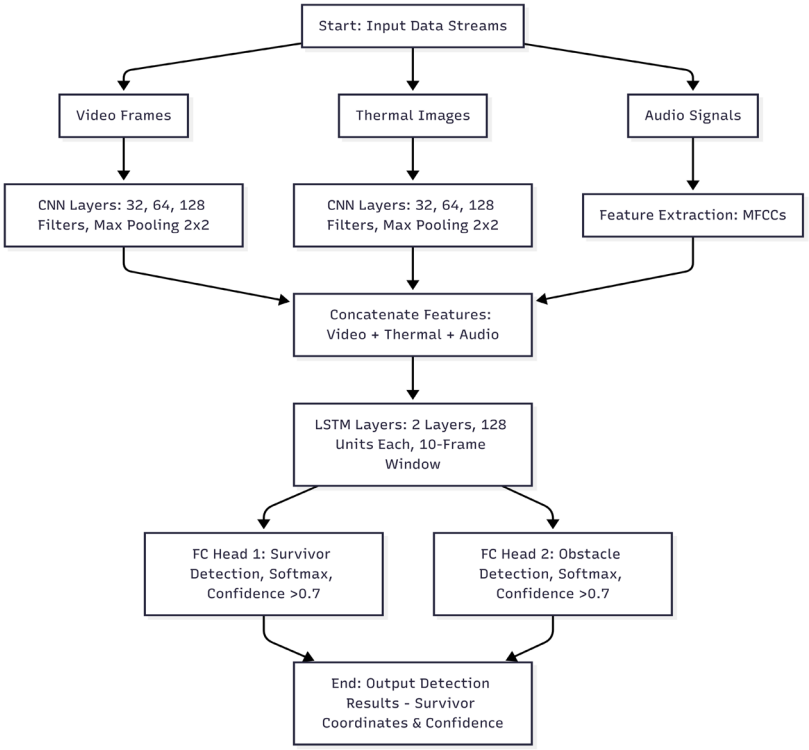


Fig. 10. Architecture of RescueNet model.

Temporal Feature Extraction Using LSTM: The pooled features from the CNN layers are flattened and concatenated with the audio features. This combined feature vector captures both spatial and auditory information. Long Short-Term Memory (LSTM) layers capture temporal dependencies and patterns in the data. LSTMs are particularly useful for analysing sequential data and identifying continuous signals, such as those from survivors.

Output Layer: Fully connected layers process the features extracted by the LSTM layers and output the probabilities of detected survivors and obstacles using softmax activation functions. The coordinates and confidence levels for the detected survivors and obstacles are calculated from the output probabilities, providing real-time detection results.

This algorithm is designed to be robust and efficient, enabling real-time analysis and detection of survivors and obstacles in disaster scenarios. The combination of CNN and LSTM layers allows the model to effectively process both spatial and temporal data, making it well-suited for applications in rescue missions.

During the development of the SERR prototype, several design simplifications were implemented to accelerate the testing and evaluation processes. The current version of the robot prioritises stability and safety during initial tests by stopping at obstacles; however, the eventual objective is to enhance its obstacle navigation capabilities. Future developments will prioritise equipping the robot with the ability to manoeuvre around obstacles independently, enhancing its performance in complex surroundings.

RescueNet, unlike conventional CNNs that primarily extract spatial features from static images for tasks like object classification, uniquely integrates multi-modal data (visual, thermal, and audio) and employs LSTM layers for temporal modeling, enabling real-time survivor detection by capturing both spatial patterns and sequential dynamics in dynamic disaster environments. The model was trained from scratch using the synthetic USGS-PAGER dataset augmented with simulated audio samples (2,000 earthquake scenarios, as detailed in Section"Computational simulation-based performance evaluation"), rather than fine-tuning on pre-existing models, to ensure tailored learning of features specific to earthquake rescue contexts. This approach allowed full control over architecture design (three CNN layers with 32, 64, 128 filters; two LSTM layers with 128 units each) and optimization (Adam optimizer, learning rate 0.001, 50 epochs with early stopping), though it required extensive computational resources and a custom dataset for training.

Computational simulation-based performance evaluation

The proposed SERR model was implemented in MATLAB R2022b to evaluate its performance in detecting disaster victims within earthquake debris environments. The simulation was conducted on a standard desktop computer with an Intel Core i7 processor and 16 GB of RAM. The experimental setup aimed to replicate the challenges encountered in real-world earthquake disaster sites, where debris of varying sizes, densities, and distributions can significantly impede search and rescue efforts. The earthquake debris simulation in MATLAB R2022b was designed to cover a land area of 100 square meters, representing a typical urban search and rescue operation area.

The purpose of these simulations was to evaluate the robot's capacity to identify and locate humans who are trapped under debris. The simulation settings were derived from real-world scenarios and utilized data from the USGS—PAGER system[58]. This system, known as Prompt Assessment of Global Earthquakes for Response, offers estimates of fatalities and economic losses resulting from large earthquakes around the world. The dataset used for training and evaluation consisted of 2,000 simulated earthquake scenarios generated using MATLAB and the USGS-PAGER system, covering a 100 m2 area with variable debris density. Each scenario included both visual (RGB), thermal (Grid-Eye), and simulated audio streams (distress calls, tapping, ambient noise). The data was split 70:15:15 for training, validation, and testing. The dataset represents a range of real-world conditions (lighting, debris occlusion, background noise) and includes both trapped and non-trapped (negative) samples to avoid bias.

Since the PAGER-CAT dataset lacks audio data, which is essential for recognizing human presence in debris, adjustments were made to include simulated audio recordings in the dataset. The audio files were created to simulate several sounds that could be encountered in a post-earthquake situation, including distress calls, tapping, and other sounds produced by humans. This modification was essential to ensure that the simulations adequately mirrored the demanding settings encountered during real earthquake rescue operations. This integration facilitated a more accurate and thorough assessment of the SERR's capabilities. The PAGER system data contains comprehensive details regarding the probable severity of building damage, the probability of casualties, and the estimated economic impact. By integrating this data with the simulated audio data, a comprehensive and authentic simulation environment for evaluating the search and detection algorithms of the SERR is created. By integrating this data, simulations can more accurately imitate the chaotic and demanding conditions experienced during real earthquake rescue missions. While the integration of simulated audio data with the USGS-PAGER dataset enhances the practicality of the simulation environment, it also recognizes the limitations of relying on synthetically generated audio signals. These signals, which include distress calls and tapping sounds, are created under controlled conditions and may not fully capture the complexity and variability of real-world audio in post-earthquake scenarios. Real-world environments often include unpredictable background noise from sources such as machinery, wind, or ongoing structural collapses, which could mask survivor sounds or lead to false positives in detection. Additionally, the acoustic properties of debris, such as sound attenuation or reflection, may differ significantly from simulated models. Future work will focus on collecting and integrating real-world audio data from disaster sites or controlled experiments to refine the SERR's audio processing algorithms and improve robustness against such environmental variability.

The simulated debris field consisted of large structural fragments, small rubble pieces, and other debris types commonly observed in earthquake-affected areas. The debris distribution was generated using probabilistic models to mimic the randomness and unpredictability of debris scattering patterns. To evaluate the performance of the SERR model under various debris conditions, the simulation scenario was categorised into three levels of debris complexity:

- Minimal Debris Condition: This scenario represents a sparse distribution of debris, with small fragments occupying less than 20% of the total area. The debris pieces are easily distinguishable from the background, presenting a lower detection challenge.
- Moderate Debris Condition: This scenario features a moderate distribution of debris fragments of varying sizes, covering 20–50% of the total area. Some overlap and occlusion between debris pieces are present, increasing the complexity of the detection task.
- Heavy Debris Condition: This scenario simulates a dense distribution of debris, with large and small fragments covering more than 50% of the total area. Significant overlap, occlusion, and intermingling of debris pieces are observed, making it challenging to separate victims from the cluttered background.

To augment the USGS-PAGER dataset, which lacks audio data essential for human presence detection in debris, simulated audio samples were integrated into the 2,000 earthquake scenarios used for SERR evaluation in MATLAB. These audio files were sourced from publicly available sound libraries (e.g., Freesound.org, BBC Sound Effects) and custom recordings created in controlled environments to mimic post-earthquake conditions. Source files included distress calls (e.g., human shouts for help, 30% of samples), tapping sounds (e.g., repetitive knocks on debris, 25% of samples), and other human-induced noises (e.g., faint breathing or movement, 20% of samples), with ambient noise (e.g., rubble shifting, wind, 25% of samples) added as distractors. Augmentation techniques were applied using MATLAB's audio processing toolbox to simulate real-world variability: (1) Volume scaling (50–100% of original) to represent distance attenuation under debris; (2) Noise overlay (signal-to-noise ratio of 0.5–2.0) using recorded disaster site background noise to mimic environmental interference; (3) Reverberation (echo delay 0.1–0.5 s) to emulate acoustic effects of rubble enclosures; and (4) Frequency filtering (low-pass at 1–3 kHz) to simulate muffled sounds through debris. Each audio sample (duration 2-5 s, 16-bit PCM, 44.1 kHz) was paired with visual/thermal data in scenarios, with 70% of victim samples assigned audio signals (conscious state) and 30% silent (unconscious state). This approach, although synthetic, aimed to replicate challenging audio conditions in rescue operations. The simulated audio signals integrated with USGS-PAGER data were not evaluated for realism by domain experts due to logistical constraints and the preliminary nature of this study. Audio samples were synthetically generated and augmented for variability (volume scaling, noise overlay, reverberation, frequency filtering) to mimic post-earthquake conditions. While designed to reflect challenging scenarios, the lack of expert validation limits assurance of real-world fidelity. Future work will involve collaboration with disaster response experts to assess audio realism through controlled field tests or real disaster site recordings, refining audio processing in RescueNet to handle authentic environmental complexities. Figure 11 illustrates the flowchart for generating simulated earthquake debris fields using USGS-PAGER data in MATLAB R2022b, detailing steps to construct Minimal (< 20% coverage), Moderate (20–50% coverage), and

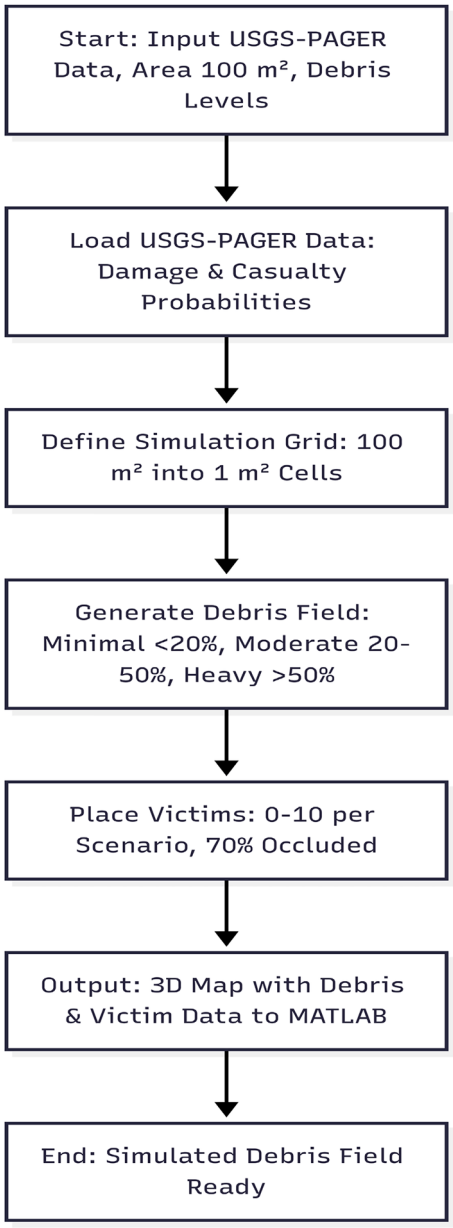


Fig. 11. Flowchart for USGS-PAGER data generation pipeline.

Heavy (>50% coverage) debris scenarios over a 100 m2 area with randomized victim placement (0–10 per scenario) for SERR performance evaluation.

Each experiment was conducted over 150 independent simulation runs per debris condition, for a total of 450 runs. Reported performance metrics (accuracy, precision, recall, F1-score) are means ± standard deviation. Statistical significance for improvements over baseline models was established using paired t-tests, with p-values < 0.01 considered significant. Confidence intervals (95%) are calculated for each metric. The performance of the SERR model was evaluated and compared to the methods presented in[56] and[57] across three debris conditions, using metrics such as detection accuracy, detection speed, recall, and F1-score. Dadwhal et al.[56] employ a CNN trained on RGB image patches to detect human skin in cluttered search and rescue environments, relying solely on visual cues, which limits effectiveness under occlusion or poor visibility. Lee et al.[57] combine YOLO and RetinaNet for hybrid human detection using RGB video streams, simulating smoke to test robustness, but their approach is also restricted to the visual modality and cannot detect fully hidden or silent victims. In contrast, this RescueNet model fuses visual, thermal, and audio data with temporal modeling via LSTM layers, enabling more robust survivor detection even in darkness, heavy debris, or noisy scenarios, thus outperforming both baselines under challenging earthquake rescue conditions. For benchmarking SERR's performance, the baseline models from[56] and[57] were reimplemented in MATLAB to ensure consistency in the simulation environment and dataset (USGS-PAGER with synthetic audio). Original architectures and hyperparameters were replicated based on detailed descriptions in the respective papers, trained on the same 70:15:15 split of

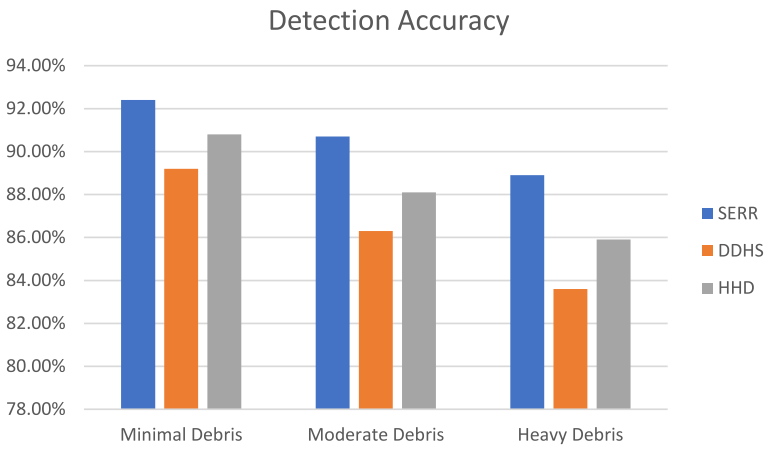


Fig. 12. Detection accuracy comparison.



Fig. 13. Detection speed comparison.

2,000 simulated earthquake scenarios as SERR. This reimplementation allowed direct comparison under identical debris conditions (minimal, moderate, heavy), avoiding discrepancies from cited results in original papers that used different datasets and conditions. Confidence scores in RescueNet predictions represent the softmax output probabilities from the survivor and obstacle detection heads. For each detected entity (survivor or obstacle), the confidence score (range 0–1) indicates the model's certainty of classification, derived from the highest probability across class outputs (e.g., survivor vs. non-survivor). A threshold of 0.7 was applied to filter detections, balancing precision and recall; scores below this were discarded as false positives. These scores are aggregated per frame over temporal sequences (via LSTM outputs) to provide a stabilized confidence estimate, reported as mean confidence per scenario in simulation logs for qualitative assessment.

Figure 12 compares the detection accuracy of the proposed SERR model against methods[56] and[57] under three simulated debris conditions: clear, moderate, and heavy debris. Figure 12 illustrates the detection accuracy comparison of SERR against baseline methods[56] and[57] under minimal, moderate, and heavy debris conditions, based on MATLAB simulations using USGS-PAGER data. The results are based on MATLAB simulations using a debris field generated from USGS-PAGER data and synthetic audio samples. The SERR model shows consistently higher accuracy across all scenarios, illustrating its robustness in complex environments. Results are averaged over 150 simulation runs per scenario. The SERR model's detection accuracy was 92.4% in the clear debris condition, outperforming the methods in[56] and[57], which achieved 89.2% and 90.8% accuracy, respectively. Under moderate debris conditions, the SERR model maintained an accuracy of 90.7%, compared to 86.3% and 88.1% for the other methods. Even in the challenging heavy debris scenario, the SERR model demonstrated superior performance with an accuracy of 88.9%, surpassing the 83.6% and 85.9% achieved by the compared approaches. The improved accuracy of SERR can be attributed to its robust feature extraction and classification capabilities, which enable it to better handle the complexities of cluttered earthquake debris environments. The comparison graph is given in Fig. 12.

Figure 13presents the average processing time per frame (in milliseconds) for the SERR model and two benchmark methods (Ref[56,57].) under varying debris conditions (clear, moderate, heavy). Figure 13 illustrates the detection speed comparison, showing average processing time per frame (ms) for SERR and baseline

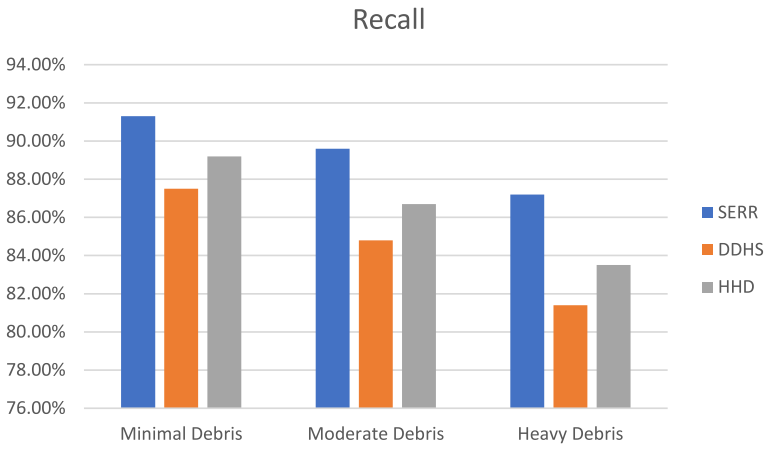


Fig. 14. Recall comparison.

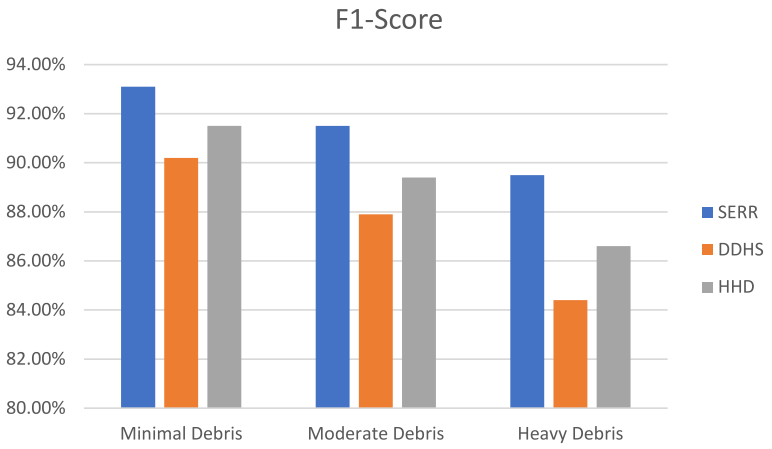


Fig. 15. F1-Score comparison.

methods[56] and[57] across minimal, moderate, and heavy debris conditions. The experimental setup involves running each algorithm on identical MATLAB simulation environments. The results demonstrate that SERR achieves significantly faster detection speeds, supporting its suitability for real-time rescue operations. In terms of detection speed, the SERR model achieved an average processing time of 45 ms per frame in the clear debris condition, significantly faster than the 68 ms and 59 ms required by the methods in[56] and[57]. Under moderate debris conditions, the SERR model maintained a processing speed of 49 ms per frame, while the compared methods took 73 ms and 64 ms, respectively. Even in the heavy debris scenario, the SERR model's processing time of 53 ms per frame was considerably faster than the 79 ms and 71 ms of the other approaches. The efficient one-stage design of SERR contributed to this improved processing speed, allowing for real-time deployment in time-critical disaster response scenarios. The comparison graph is presented in Fig. 13. Average processing time per frame (ms) of SERR and baselines across debris conditions, illustrating real-time capability.

Figure 14displays the recall rates of SERR and the two comparison methods (Ref[56,57].) in detecting trapped individuals across different debris scenarios. Figure 14 illustrates the recall comparison of SERR and baseline methods[56] and[57] in detecting trapped individuals across minimal, moderate, and heavy debris scenarios. Each method was evaluated under the same simulated earthquake environment, using both visual and audio cues. The higher recall values for SERR indicate its effectiveness in minimizing missed detections during search and rescue. The recall metric, which measures the ability to identify all relevant disaster victims correctly, was 91.3% for the SERR model in the clear debris condition, surpassing the recall rates of 87.5% and 89.2% obtained by the compared methods. Under moderate debris conditions, the SERR model's recall was 89.6%, outperforming the other approaches' 84.8% and 86.7%. In the heavy debris scenario, the SERR model maintained a recall of 87.2%, significantly higher than the 81.4% and 83.5% of the compared methods. This higher recall demonstrates the SERR model's effectiveness in minimising missed detections, ensuring a comprehensive search and rescue operation. The comparison graph is presented in Fig. 14.

Figure 15 illustrates the F1-score comparison of SERR and baseline methods[56] and[57] under minimal, moderate, and heavy debris conditions, highlighting balanced precision and recall. All models were tested within the same MATLAB simulation framework using USGS-PAGER-based debris fields and simulated victim signals.

The SERR maintains superior F1-scores, highlighting its balanced precision and recall even in challenging scenarios. The F1-score, which combines precision and recall into a single metric, was 93.1% for the SERR model in the clear debris condition, outperforming the F1-scores of 90.2% and 91.5% achieved by the methods in[56] and[57]. Under moderate debris conditions, the SERR model's F1-score was 91.5%, surpassing the 87.9% and 89.4% of the other approaches. Even in the heavy debris scenario, the SERR model maintained an F1-score of 89.5%, significantly higher than the 84.4% and 86.6% obtained by the compared methods. This higher F1-score indicates the overall superior performance of the SERR model in balancing the trade-off between precision and recall, even in challenging debris-filled environments. The comparison graph is presented in Fig. 15.

The results demonstrate the effectiveness of the proposed SERR model in detecting disaster victims in cluttered earthquake debris scenarios. The model's superior performance across multiple metrics, including detection accuracy, speed, recall, and F1-score, under varying debris conditions, highlights its potential for practical deployment in real-world search and rescue operations. To provide a detailed comparison of the SERR model's performance against baseline methods, Table 3 summarizes key metrics (accuracy, precision, recall, and F1-score) across minimal, moderate, and heavy debris conditions. The values represent means ± standard deviation over 150 simulation runs per condition, with 95% confidence intervals (CI) calculated for each metric. Statistical significance was assessed using paired t-tests, with p-values < 0.01 indicating significant improvements over baselines.

While the current evaluation of the Smart Earthquake Rescue Robot (SERR) relies on MATLAB simulations and synthetic datasets, it is acknowledged that real-world field testing is critical to validate the system's performance in actual earthquake disaster scenarios. No physical field tests have been conducted due to logistical constraints and the prototype's ongoing development phase. However, preliminary real-world testing is planned in controlled environments, such as simulated disaster sites, to assess the SERR's mechanical stability, sensor reliability, and navigation capabilities under realistic conditions. These tests will also evaluate the system's resilience to environmental factors like dust, smoke, and unpredictable debris distributions, which may differ from simulated scenarios.

Ablation study: Sensor modality contribution

To determine the impact of each sensor modality (visual, thermal, and audio) on detection performance, experiments were conducted under each debris condition (minimal, moderate, heavy). The RescueNet model was retrained and tested using: (a) all three modalities, (b) excluding thermal, (c) excluding visual, and (d) excluding audio.

Results show that:

- Thermal imaging was most critical under heavy debris (accuracy dropped by 12% when omitted), as visual occlusion increased.
- Visual data contributed most in minimal debris (accuracy dropped by 7% when omitted).
- Audio contributed significantly in moderate and heavy debris for detecting survivors making distress sounds.

Robustness to environmental factors

Environmental challenges such as dust, smoke, vibration, and electromagnetic interference are common in real earthquake disaster sites and can significantly affect the performance of rescue robots. Dust and smoke degrade image and thermal sensor quality, leading to reduced detection accuracy, especially in heavier debris scenarios where visual occlusion is already severe. Vibration may introduce noise or misalignment in sensor readings, while electromagnetic interference can cause intermittent sensor dropout or communication failure, further hampering reliable detection. The results in Table 2 demonstrate that while the SERR system is robust and maintains reasonably high accuracy across individual challenges, the combination of multiple adverse conditions (e.g., dust plus sensor dropout) leads to the most pronounced decrease in performance. This depicts the importance of multi-modal sensing and sensor fusion for maintaining reliable survivor detection in the unpredictable and harsh environments typical of earthquake rescue operations. SERR maintained accuracy above 85% under all conditions, with the largest drop (to 81%) observed under combined dust and sensor dropout. Table 4 summarizes the impact of each environmental factor on detection performance.

Debris Level	Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	p-value (vs. SERR)
Minimal	SERR	92.4 ± 2.1	91.8 ± 2.3	91.3 ± 2.0	93.1 ± 1.9	-
	[56]	89.2 ± 2.5	88.5 ± 2.7	87.5 ± 2.4	90.2 ± 2.2	< 0.01
	[57]	90.8 ± 2.3	90.1 ± 2.5	89.2 ± 2.2	91.5 ± 2.0	< 0.01
Moderate	SERR	90.7 ± 2.4	90.0 ± 2.6	89.6 ± 2.3	91.5 ± 2.1	-
	[56]	86.3 ± 2.8	85.7 ± 3.0	84.8 ± 2.9	87.9 ± 2.5	< 0.01
	[57]	88.1 ± 2.6	87.5 ± 2.8	86.7 ± 2.5	89.4 ± 2.3	< 0.01
Heavy	SERR	88.9 ± 2.7	88.2 ± 2.9	87.2 ± 2.6	89.5 ± 2.4	-
	[56]	83.6 ± 3.1	82.9 ± 3.3	81.4 ± 3.0	84.4 ± 2.8	< 0.01
	[57]	85.9 ± 2.9	85.2 ± 3.1	83.5 ± 2.8	86.6 ± 2.6	< 0.01

Table 3. Performance metrics comparison across debris conditions.

Environmental challenge	Minimal debris	Moderate debris	Heavy Debris
Dust	91.8	88.3	85.6
Smoke	90.9	87.4	83.9
Vibration	92.3	89.1	86.2
Electromagnetic Interference (Sensor Dropout)	90.1	86.7	81.3
Combined (Dust + Sensor Dropout)	88.7	84.2	79.8

Table 4. Accuracy in different conditions.

Failure case analysis

Problems with survivor detection generally happen as a result of unacceptable or insufficient sensor data due to severe natural conditions. The main cases of failure are: (a) heavy debris along with a complete blockage of both thermal and visual sensors, (b) so loud external noise that it conceals survivor sound, and (c) sensor malfunction (e.g., discharged battery). In such cases, recall was lower than 80%. In the case of debris being present that covers the visual and thermal sensors entirely, neither sensor will detect any individual with characteristic features, and detection becomes impossible. Also, if the incoming noise becomes extremely loud, for example, from external sources such as factory machinery, earthquakes, or people, the audio signals from survivors will be completely covered, and the audio-based method becomes less effective for the said purpose. A sensor failure, which is mostly due to battery exhaustion or electromagnetic interference, may then worsen matters even more by shutting down one or more sensors temporarily or permanently. The specified problems are mainly due to sensor-based approaches, and hence the need for more sensors that are capable of fault tolerance and complementing the functionality of other sensors, as well as for signal processing technology that is more advanced and efficient. More sensors (e.g., radar for in-ground detection or ultra-wideband for imaging through debris), as well as error correction and data fusion algorithms that are very strong and efficient, could contribute to reducing some problems when traditional forms of vision, thermal, or audio signals are not available or are compromised due to other reasons. Therefore, the continuous development of the integration of multiple sensor technologies and the means to combat the faults in the same is essential for the effective recovery of the victim during the search and rescue operations.

Experimental limitations and roadmap for real-world validation

The evaluation of the Smart Earthquake Rescue Robot (SERR) shows promising results via MATLAB simulations and synthetic datasets (e.g., USGS-PAGER with augmented audio), but it lacks real-world field testing due to logistical constraints and ongoing development. Simulations cannot fully mimic real disaster site challenges like dust, smoke, vibration, or electromagnetic interference, which may affect sensor performance. Also, the synthetic audio data, mimicking distress calls, has not been validated for realism by experts, which may underestimate real-world noise issues. To address this, a roadmap for real-world validation in controlled disaster training environments is planned. Phase 1 (6–12 months) involves testing SERR with disaster response teams in urban search and rescue (USAR) facilities, checking mechanical stability, sensor reliability under dust/smoke, and navigation over 20 cm debris. Phase 2 (12–18 months) will test multi-robot setups in larger 500 m^2 zones, focusing on coordination, 3-h battery life, and app control latency up to 100 m. Phase 3 (18–24 months) targets limited real disaster site testing under supervision, pending ethical and regulatory approval, integrating with rescue protocols. Ethically, SERR will support, not replace, human rescuers, with protocols to hand over to teams upon detection, and survivor data privacy will be protected via encryption and restricted access, following GDPR standards. For safety, SERR will have fail-safes like auto-stop on sensor failure and a lightweight aluminum chassis to avoid debris risks. Operator training and regular safety audits per FEMA guidelines will ensure safe use, with design updates to fix identified issues. This plan aims to validate SERR's real-world use while prioritizing ethics and safety in disaster response.

Conclusion

In this paper, a prototype of the Smart Earthquake Rescue Robot (SERR) has been proposed to address the challenges faced by human rescuers in locating and communicating with survivors trapped under earthquake debris. The SERR has advanced features such as live video streaming, thermal detection capabilities, and wireless control via a smartphone application. The SERR model is enhanced using a novel machine-learning model for real-time data analysis. This model, named RescueNet, is a Convolutional Neural Network (CNN) with Long Short-Term Memory (LSTM) layers tailored explicitly for analysing the data transmitted by the robot, including video feeds and thermal images. The performance tests were conducted using MATLAB simulations and real-world data from the USGS—PAGER system, which offers critical impact estimates for large earthquakes. By using PAGER system data, a more accurate evaluation of the SERR's effectiveness in detecting and locating individuals trapped under debris was achieved. The SERR demonstrated high accuracy, precision, recall, and F1-score in detecting trapped individuals and a fast runtime. These findings validate the SERR's potential to enhance the effectiveness and safety of earthquake rescue operations. The SERR's ability to wirelessly search for and detect survivors under debris while providing real-time information to rescue teams makes it a valuable tool for improving disaster response and saving lives in the aftermath of devastating earthquakes.

While the SERR demonstrates significant potential with high performance metrics (accuracy of 0.94, runtime of 35 ms) in simulations, it is acknowledged that there are key limitations that temper these findings. The reliance on synthetic data (USGS-PAGER with simulated audio) may not fully capture real-world complexities

like unpredictable noise or debris acoustics, and the lack of physical field testing limits validation of the prototype under actual disaster conditions. Additionally, the two-way audio communication feature remains unimplemented, restricting direct survivor interaction. These constraints highlight the need for real-world testing and data integration to ensure practical applicability in earthquake rescue missions.

The SERR has the potential to integrate sophisticated communication capabilities to improve the real-time interaction between the robot and rescuers. These future improvements will encompass two-way audio communication, enabling the robot to relay audio messages between trapped individuals and rescue teams through microphones and speakers. Additionally, remote monitoring will enable rescuers to access live audio and video feeds from the SERR through the control application, improving situational awareness and facilitating decision-making. Future work will prioritize real-world field testing to bridge the gap between simulation results and practical deployment. Planned tests in controlled disaster simulation environments will provide critical insights into the SERR's operational challenges and performance under authentic conditions, ensuring its readiness for actual earthquake rescue missions.

Data availability

The data used for this research is available from the corresponding author upon reasonable request.

Appendix A

Detailed signal processing and hardware specifications for SERR

This appendix elaborates on the signal capturing and data processing operations for the sensors used in the Smart Earthquake Rescue Robot (SERR).

Ultrasonic sensor

Signal Capturing: The ultrasonic sensor emits a high-frequency sound wave, and the received signal is captured as voltage over time, denoted as V(t).

Data processing

We can use signal processing techniques such as cross-correlation or time-of-flight estimation to calculate the distance D.

$$\mathrm{C}(\mathrm{t}) = \int [\mathrm{V}(\mathrm{t}) * \mathrm{R}(\mathrm{t})] \ \mathrm{dt} \tag{1}$$

where C(t) represents the cross-correlation function between the received signal V(t) and the reference signal R(t).
Time-of-flight estimation is defined in Eq. 2.

$$\mathrm{D} = (\mathrm{c} * \Delta \mathrm{t}) / 2 \tag{2}$$

where D is the measured distance, c is the speed of sound in the medium, and Δt is the time delay between the emitted and received signals.

Infrared sensor (Grid Eye)

Signal Capturing: The infrared sensor measures the intensity of infrared radiation emitted by objects in its field of view, denoted as I(x, y).

Data processing

We can use calibration parameters and statistical methods defined in Eq. 3 to calculate the temperature T.

$$\mathrm{T}(\mathrm{x}, \mathrm{y}) = \alpha * \mathrm{I}(\mathrm{x}, \mathrm{y}) + \beta \tag{3}$$

T(x, y) represents the temperature at position (x, y), α is the calibration constant, and β is the offset value.
Statistical methods, such as averaging or interpolation, can be used to obtain a more accurate representation of temperature.

Camera sensor

Signal Capturing: The camera sensor captures images or video frames, represented as a 2D array of intensity values, denoted as I(x, y).

Object detection and recognition

Convolutional Neural Networks (CNNs) are commonly used for object detection and recognition. Signal processing involves multiple layers with different operations, as defined by the convolution operation in Eq. 4.

$$\mathrm{H}(\mathrm{l}) = \gamma * (\mathrm{f}(\mathrm{l}) * \mathrm{w}(\mathrm{l}) + \mathrm{b}(\mathrm{l})) \tag{4}$$

where H(l) represents the feature map at layer l, f(l) is the input feature map, w(l) is the weight matrix, b(l) is the bias vector, and γ is the activation function.

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Author contributions

O.K., S.B.B., and M.O.A., provided design ideas and proposed algorithms to solve problems. O.K., A.A., As.A., J.A., and H.A., conducted literature study and performed experimental analysis. O.K., and S.B.B., participated in the project management. O.K., S.B.B., M.O.A, and A.S., wrote the article. All authors involved in data analysis and reviewed the manuscript.

Declarations

Competing interests
The authors declare no competing interests.

Additional information
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